Unit 4 :: Data Preparation and Management

SEC-03-PHY-91-R-03

Dr. Parag Bhattacharya
Department of Physics
Rangapara College (Autonomous)

Syllabus

Unit 4: Data preparation and management

Basic pre-processing of data: filtering, sorting, normalization and handling missing values; managing multisheet workbooks and structured datasets, importing data from diverse file formats (CSV, ASCII, DAT), and ensuring interoperability between different spreadsheet and analysis platforms.

1 Importing data from diverse file formats

Suppose a physics experiment records a variable which is of interest to us, such as temperature, pressure, voltage, count rate etc. These variables are recorded (or sampled) at a sequence of time instants. Now this raw record of data produced by an instrument is almost certainly unfit for direct analysis because

a) It may contain unreliable readings.

b) It may be out of chronological order.

c) It may mix standard and non-standard units.

d) It may contain gaps.

Then we say that such a dataset must be *pre-processed* before it can be visualised or fitted to a theoretical curve.

For the purpose of pre-processing, the dataset must be first stored and archived in the form of a file. Three file formats are widely physics laboratory exports:

1. Comma-separated values (CSV files)
2. Instrument-specific data (DAT files)
3. Microsoft Excel spreadsheets (XLSX files)

1.1 ASCII files

ASCII, which is an acronym for *American Standard Code for Information Interchange*, is a character encoding standard. Originally, it was used for representing a particular set of 95 printable characters and 33 control characters, thereby giving a total of 128 code points, using 7 bits. Later, an 8-bit encoding extended the ASCII set to a total of 256 characters.

An ASCII file is one that contains data made up of ASCII characters, i.e. it is essentially raw text. Each byte in the file contains one character that conforms to the standard ASCII code.

Source codes of computer programs, batch files, macros and scripts are straight text and stored as ASCII files. HTML and XML files are also ASCII files. Text editors such as Notepad create ASCII files as their native file format.

CSV and DAT files are two of the most commonly used ASCII file formats used for storing scientific data for subsequent analysis.

1.1.1 CSV files

A comma-separated values (CSV) file is a plain text data format for storing tabular data where the fields (values) of a record are separated by a comma and each record is a line (i.e. newline separated). For example, the following are the contents of a CSV file:

```
id, name, email
1, John, john.doe@example.com
2, Jane, janey72@test.org
```

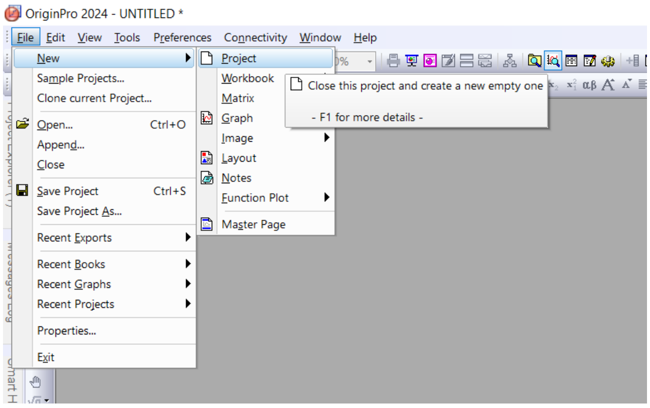
In software that generally deals with tabular data such as a database or a spreadsheet, the contents of the above CSV file is parsed in the form of a table as follows:

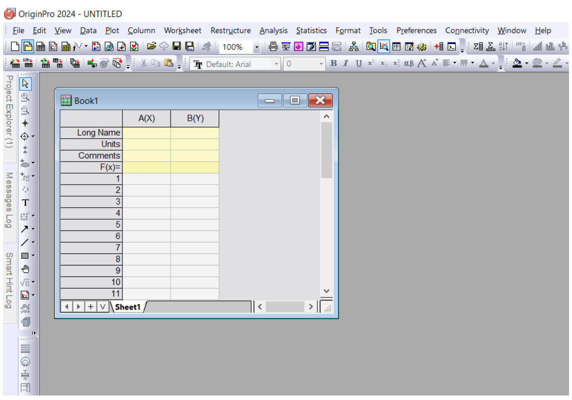
id	name	email
1	John	john.doe@example.com
2	Jane	janey72@test.org

The first line is used as the table header and the remaining lines are used as the rows of the table.

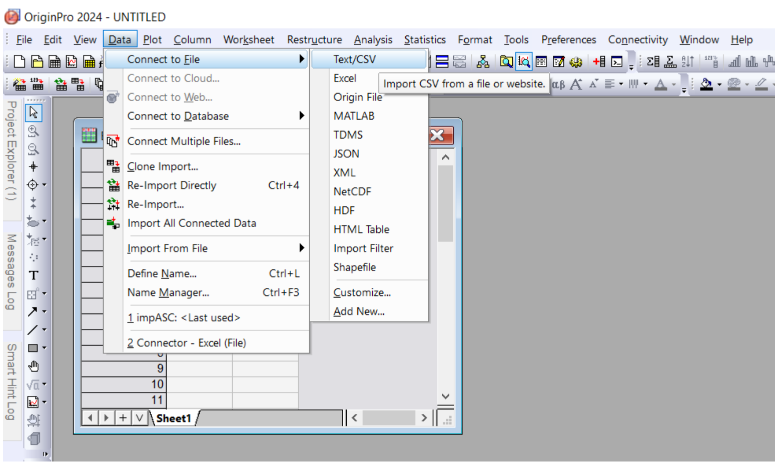
Importing data from a CSV file in Origin:

I. In the main menu of Origin, go to `File` > `New` > `Project`. This would create a new Origin project with an empty worksheet.

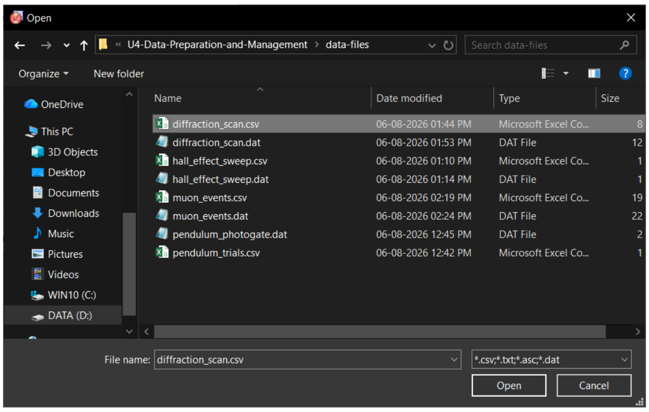




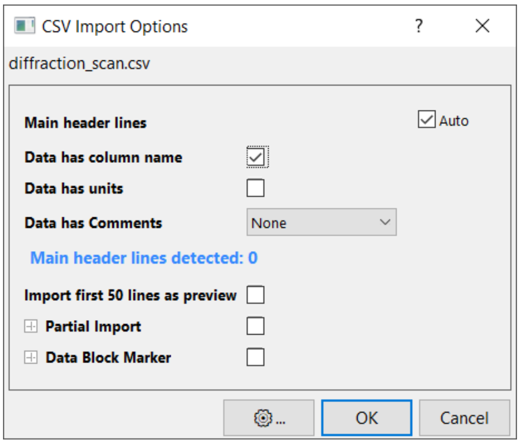
II. To fill in the worksheet with values from the CSV file, we go to `Data` > `Connect to file` > `Text/CSV`.

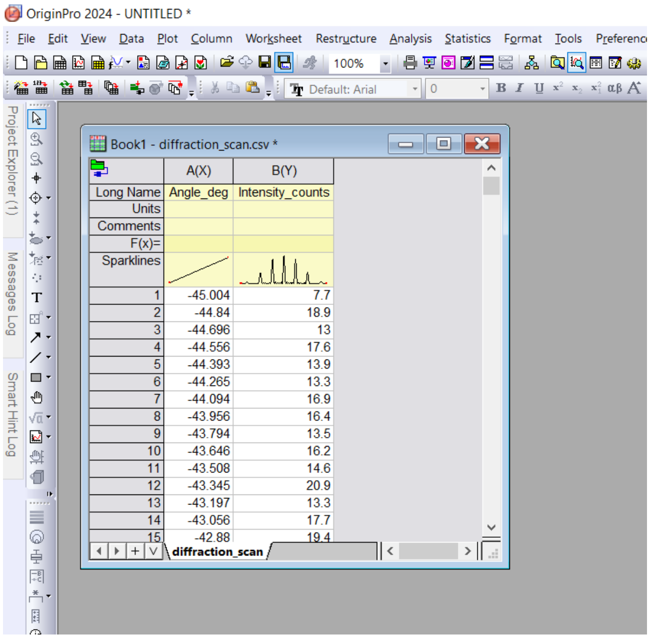


III. In the file open dialog, we navigate to the folder containing the CSV file. Selecting the file of interest, we click `Open`.



IV. Another dialog appears where we can provide the options for importing data from the CSV file. Keeping everything unchanged, we click `OK`. We find that the data from the CSV file has been imported into the worksheet and is ready for further analysis.





1.1.2 DAT files

Computer programs create DAT files to store specific information or data. The information in the file is only relevant to the program that created it, though almost any program can create DAT files. DAT might look like an acronym, but the extension name is actually short for 'data.'

A DAT file may begin with header lines that begin with #. Any line that begins with a # is treated like a comment and is subsequently ignored. So we can make use of these header lines to insert specific information in the DAT file. However, it is possible to have DAT files without such header lines.

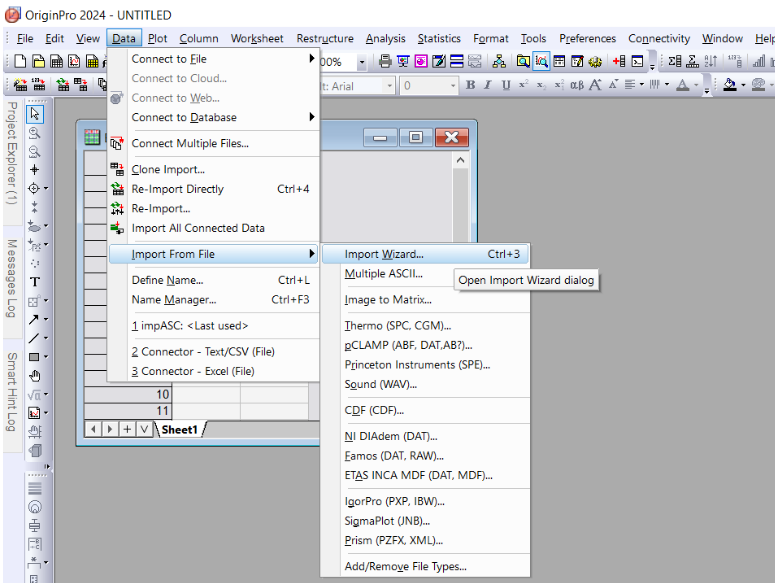
The actual data in a DAT file may be entered with each record in its own row and separated by whitespaces, such as space or tab. For example, the following are the contents of a DAT file:

```
1  # Diffraction Grating Intensity Scan
2  # Wavelength_nm: 632.8
3  # Grating_spacing_nm: 3000.0
4  # Slit_width_nm: 800.0
5  # Date: 2026-08-01
6  Angle        Intensity
7  degree       counts
8  -45.000      10.5
9  -44.850       9.6
10 -44.700      16.3
11 -44.550      11.5
12 -44.400      15.2
```

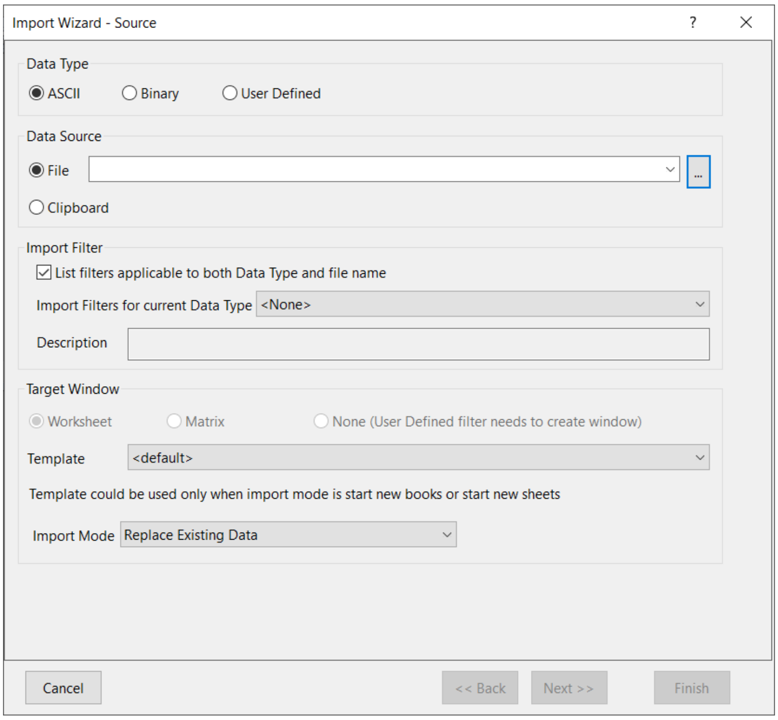
In the above DAT file, lines 1 to 5 are indicated as headers. Lines 6 and 7 are provide the column labels and units respectively. The subsequent lines contain the data, separated by a tab space.

Importing data from a DAT file in Origin:

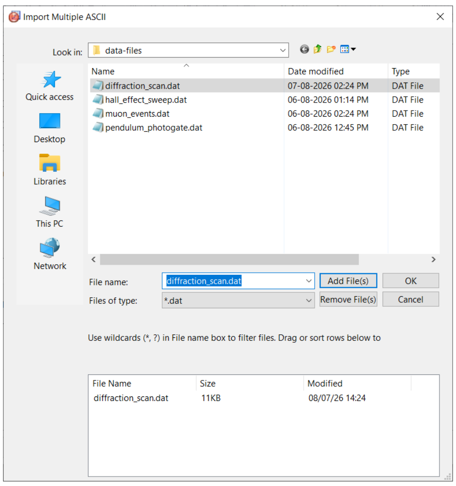
I. Create a new project in Origin. Then to import data from a DAT file, go to `Data` > `Import From File` > `Import Wizard`.

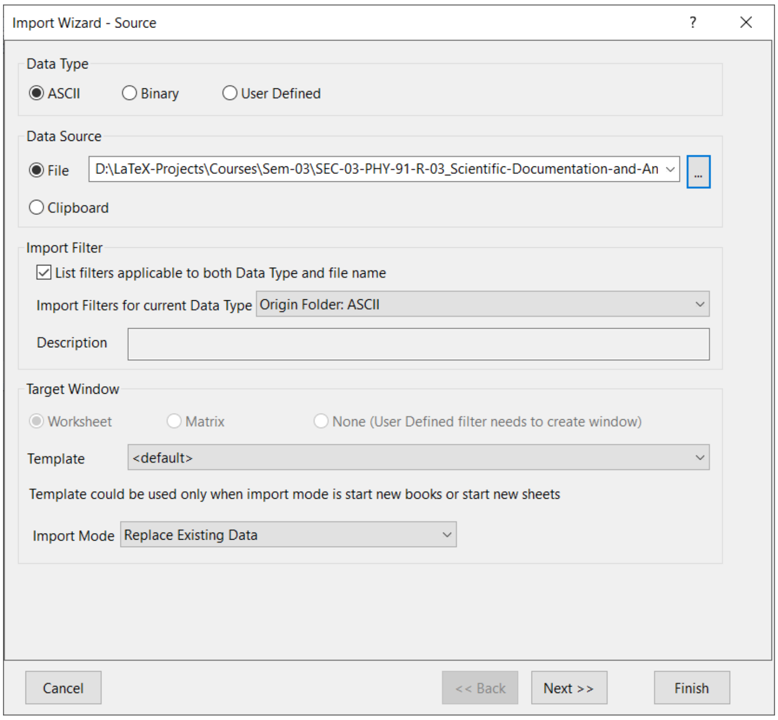


II. In the Import Wizard dialog box, we ensure that under `Data Source`, the `File` option is checked and then we click on the button to the right of the filename textbox.

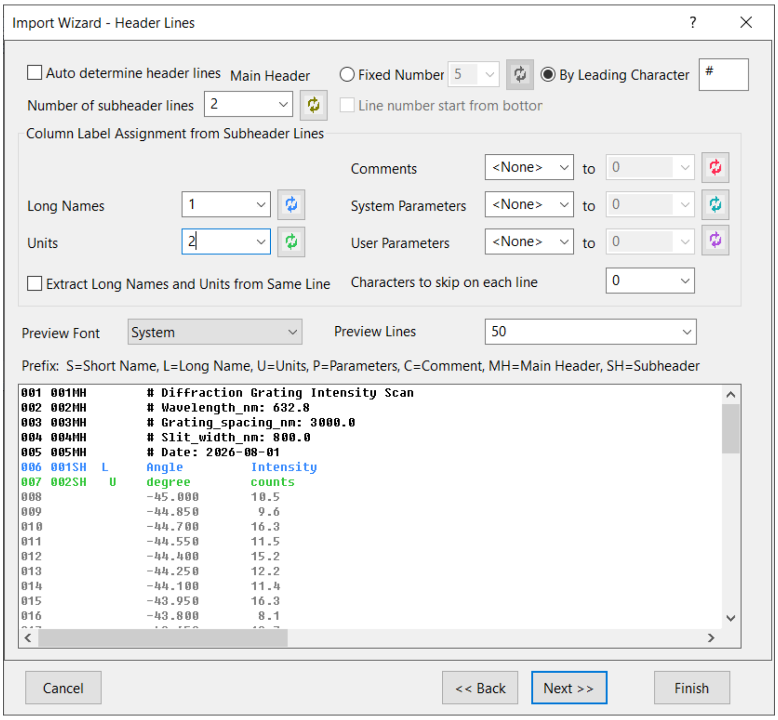


III. After navigating to the folder that contains the DAT file of interest, we select the file (or multiple files). Then we click `Add File(s)` and finally `OK`. We would be taken back to Import Wizard dialog where we can confirm that the full path of the selected DAT file is displayed in the filename textbox.

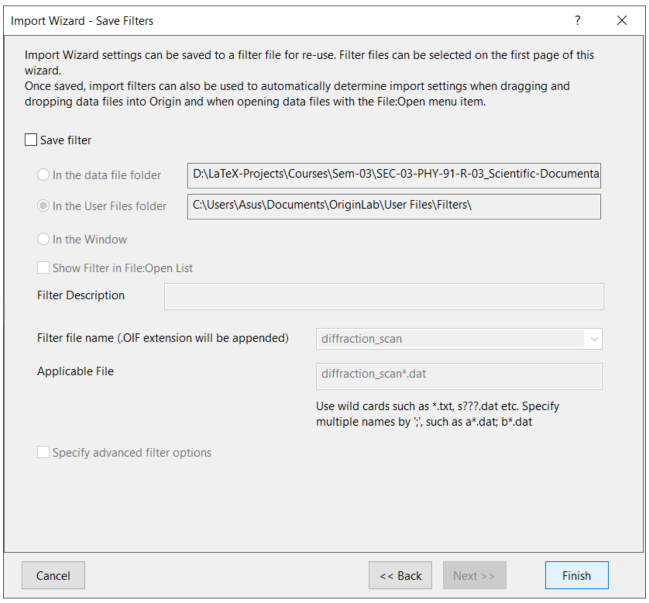




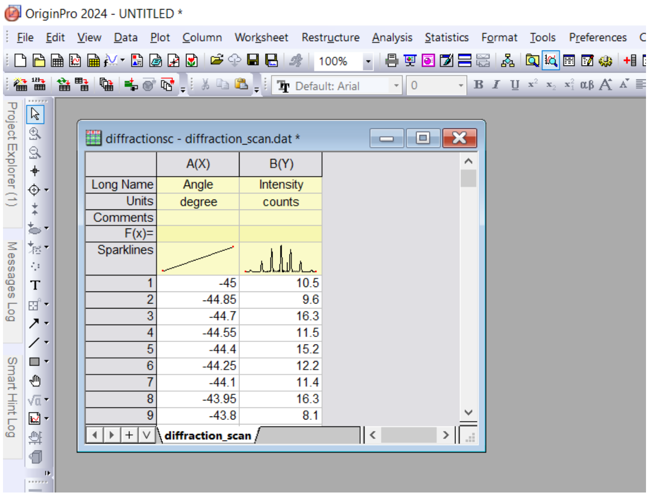
IV. Clicking `Next` on the Import Wizard dialog takes us to the next window where we set the options for data import.
We uncheck the `Auto determine header lines` checkbox. Then we enable the `By Leading Character` radio button and enter # in the textbox alongside. This is because in our DAT files, the header lines are indicated by the character # at the beginning.
We enter the input `Number of subheader lines = 2` because two lines have special meanings as containing the long names of the columns and their units. We indicate these by the inputs `Long Names = 1` and `Units = 2`.
A preview of the DAT file at the bottom of the dialog page confirms that our options are correct by displaying the header lines in black, the long names in blue, the units in green and the data in grey.



V. We click on `Next` for the remaining dialog pages and finally click on `Finish`.



VI. When the Import Wizard closes, we find that the data from the DAT file has been correctly imported into the Origin workbook and it is now ready for analysis.



1.2 Interoperability using spreadsheets

1.2.1 XLSX files

An XLSX file is the native workbook format of Microsoft Excel. It is also read and written compatibly by most other spreadsheet platforms, such as LibreOffice Calc and Google Sheets.

Structurally, an XLSX file is fundamentally different from the CSV and DAT files because it is not an ASCII file. Rather than being a flat sequence of human-readable characters, an XLSX file is a compressed (ZIP) container holding several internal XML documents that together describe one or more sheets, along with their formatting, formulas, and metadata. Hence, it cannot be usefully opened in a plain text editor the way a CSV or DAT file can.

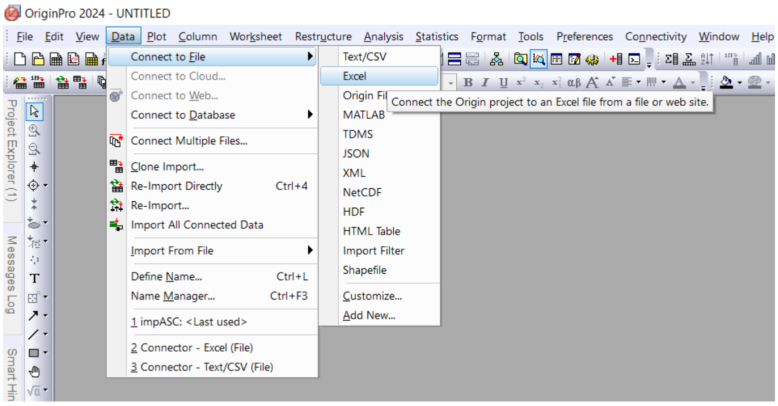
A CSV or DAT file can be parsed by any program capable of reading text, provided the delimiter and header conventions are known. An XLSX file, by contrast, can only be read by software that implements the Office Open XML specification (or a compatible library); Origin does this via a dedicated Excel import module rather than the ASCII import wizard.

A single XLSX workbook may contain multiple sheets, each an independent two-dimensional grid of cells. Unlike an ASCII file, a sheet can, by itself, carry more than one header row without any special import configuration. For example, a column-name row followed by a units row.

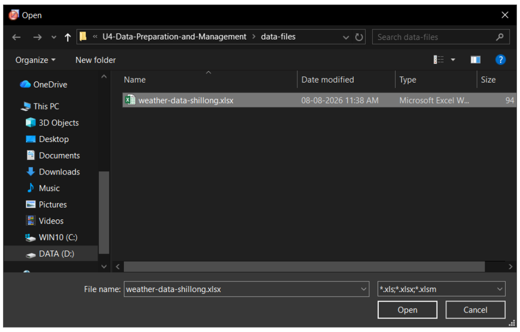
1.2.2 Working with XLSX files in Origin

Importing data from an XLSX file

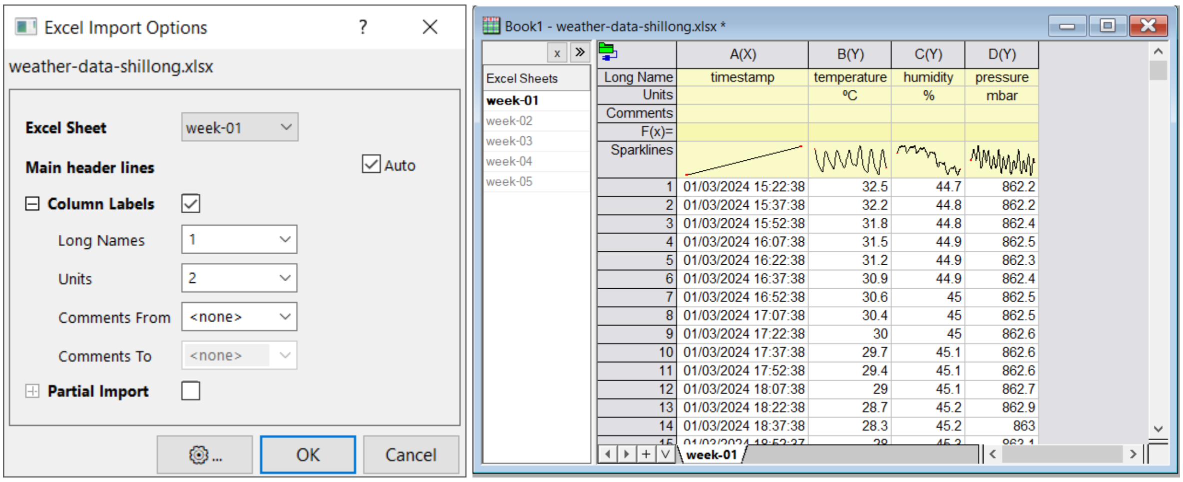
I. Create a new project in Origin. Then to connect to an XLSX file, go to `Data` > `Connect to File` > `Excel`.



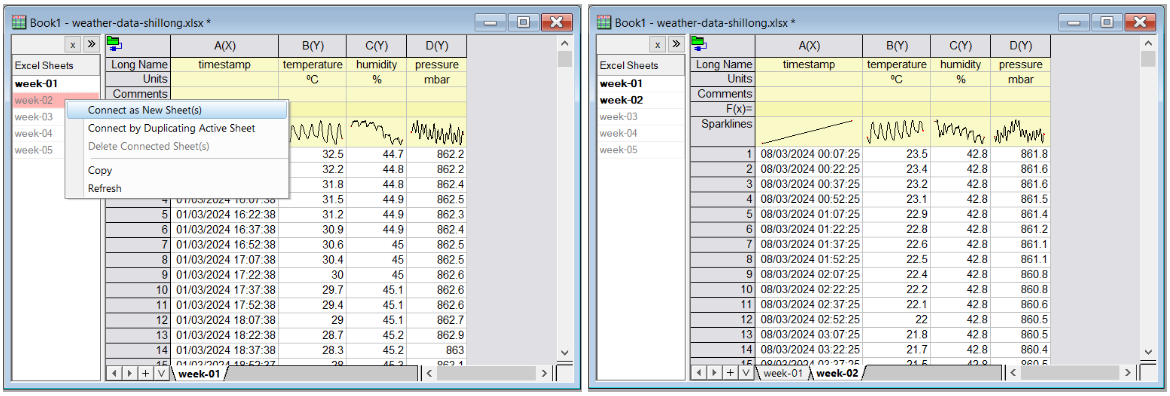
II. After navigating to the folder that contains the XLSX file of interest, we select the file and then click `Open`.



III. In the `Excel Import Options` dialog, we set `Excel Sheet` as any one of the sheets, usually the first. Also we set `Long Names = 1` and `Units = 2`. Then on clicking `OK`, we find that the data from the selected sheet has been imported into the workbook. However, we also find that, in the `Excel Sheets` sidebar, the other sheets are listed, although they are greyed out, which means that their data has not been imported.



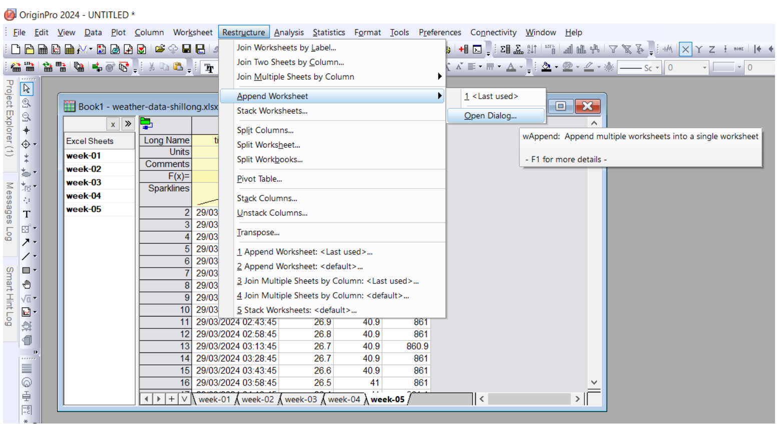
IV. To import data from another sheet of the XLSX file, we select that sheet from the sidebar, then right-click and select `Connect as New Sheet(s)`. This will import its data into a new worksheet.



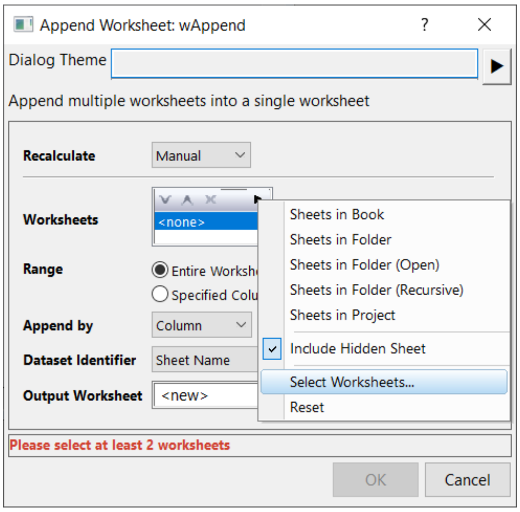
Likewise, we can import data from all the remaining sheets into separate worksheets.

Appending data from multiple sheets

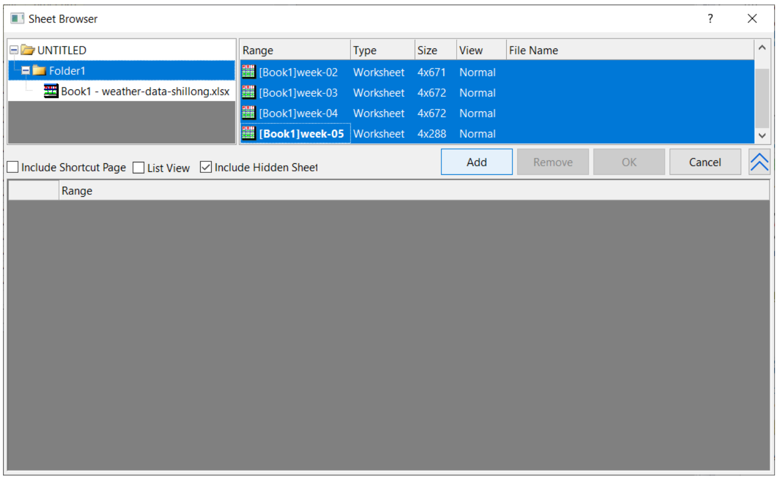
I. Suppose that we have similar data in the different sheets of an XLSX file that we have imported. Now to combine the data into a new sheet, we go to `Restructure` > `Append Worksheet` > `Open Dialog`.



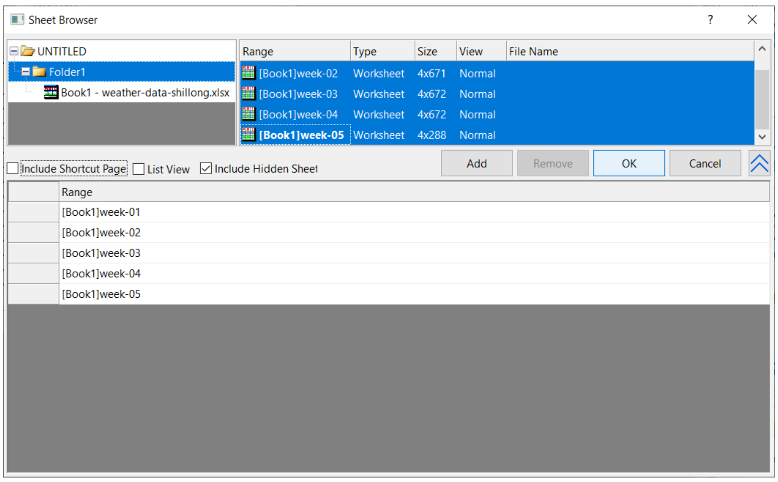
II. On the `Append Worksheet` dialog, click on the triangle button to the right of the `Worksheets` option and then
click `Select Worksheets`.



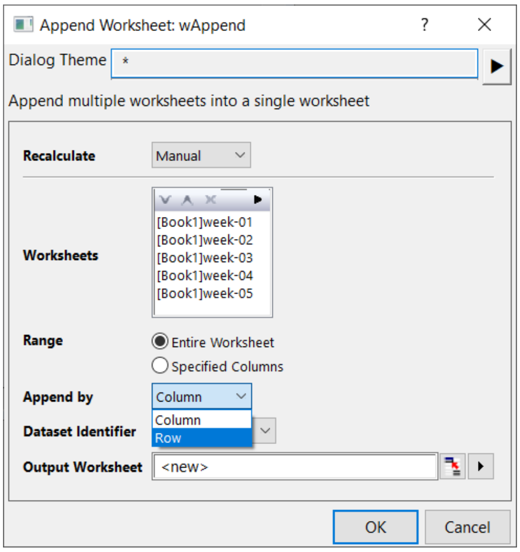
III. The `Sheet Browser` dialog would then open up. Pressing and holding down the Ctrl button, click and select all the worksheets, and then click `Add`.



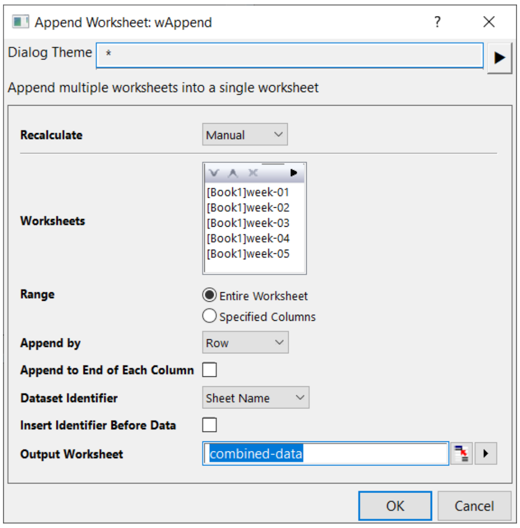
IV. The selected worksheets would then appear in the list below. We then click `OK`.

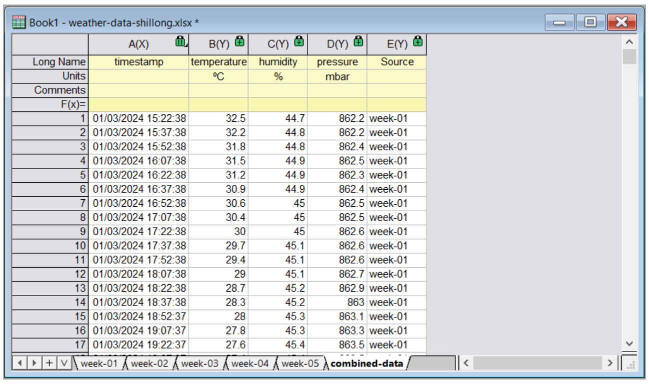


V. After the above step, we would be back in the `Append Worksheet` dialog. We set `Append by = Row`.



VI. In `Output Worksheet` we type the name of the new worksheet which is to contain the combined data. Then we click `OK` to find that a new worksheet has been created, which contains the combined data from all the worksheets.





2 Masking and filtering

In physical laboratory experiments, data recorded by instruments often contains unwanted noise, electrical spikes, or out-of-range readings. Filtering and masking are two fundamental techniques used to clean data before plotting or mathematical fitting.

2.1 Masking data

In computational data analysis software like Origin, masking operates at the data cell or point level by attaching an internal metadata boolean flag (a mask status bit) to specific entries within a data structure. The raw numerical array remains entirely intact in memory. Masking does not alter, reindex, or delete row entries from the underlying data vector.

When statistical routines (such as mean, variance, or polynomial fitting) or graphic rendering engines query the dataset, they check the mask attribute of each cell. If a cell's mask flag is evaluated as `TRUE` (active mask), the processing engine skips that memory address during iteration or sets its inclusion weight to zero.

Masking allows arbitrary cell-level selection regardless of contiguous row order. On the user interface layer, the software highlights masked cells with an overlay color (typically red) and excludes them from graph series without breaking the continuity of the remaining row index mappings.

2.1.1 Masking points in Origin

I. Open the worksheet containing the dataset.

II. Highlight the specific cells or rows that are to be masked and do any of the following:

  (a) Right-click the highlighted selection and click `Mask` > `Apply`.

  (b) From the main menu, go to `Column` > `Mask` > `Apply`.

  (c) On the shortcut menu, toggle the `Mask/Unmask` button.

  (d) Click on the `Mask Range` button on the mask toolbar.

III. The background of the selected cells turns red. The colour of the masked cells can be changed by clicking on the `Change mask color` button on the toolbar.

IV. Any plot created from this dataset will indicate the masked cells in the same colour. The masked data points can also be hidden from the plot by toggling the `Hide/Show masked points` button on the toolbar.

V. To remove the mask, highlight the masked cells and do any of the following:

  (a) Right-click the highlighted selection and click `Mask` > `Remove`.

  (b) From the main menu, go to `Column` > `Mask` > `Remove`.

  (c) On the shortcut menu, toggle the `Mask/Unmask` button.

  (d) Click on the `Unmask Range` button on the mask toolbar.

2.2 Filtering data

Filtering operates as a relational query or dynamic predicate evaluation applied across entire records (rows) of a dataset based on logical conditions defined for one or more variables.

The software evaluates a conditional logical expression, such as $f(x_i) = \texttt{TRUE}$ for $x_i \geqslant x_{\text{threshold}}$, across a specified column vector $\mathbf{x}$. This produces a global boolean mask vector corresponding to all rows in the worksheet.

Filtering creates a subset view (a temporary relational selection) of the worksheet. Rows where the condition evaluates to `FALSE` are collapsed or hidden in the user interface and bypassed by downstream analysis routines.

Unlike cell-level masking, filtering acts on full rows across all columns simultaneously to preserve tabular record integrity. When the filter rule is toggled off or modified, the full dataset view is restored instantaneously because no data points were modified or unlinked.

2.2.1 Filtering points in Origin

I. Open the worksheet containing the dataset.

II. Click on the column on which the filter is to be applied and do any of the following:

   (a) From the main menu, go to `Column` > `Filter` > `Add or Remove Filter`.

   (b) Click on the `Add/Remove data filter` button on the worksheet data toolbar.

III. Once a filter has been applied to a column, click on the filter icon on the top-left of the column header and click on the appropriate option from the context menu to modify the filter.

IV. A filter can be enabled or disabled by selecting the filtered column and clicking on the `Enable/Disable data filter` button on the worksheet data filter.

V. To remove a filter, click on the filter icon on the column header and then click `Clear Filter`.

Exercises

1. Import `muon_events.csv` into a fresh worksheet. Apply a data filter on `Pulse_Height_mV` to display only high-energy events where the pulse height is strictly greater than 50.0 mV. How many total rows remain visible after applying this filter?

2. Import `diffraction_scan.csv`. In optical diffraction measurements, detector saturation occurs at an intensity of 4095.0 counts. Identify the saturated rows around 0° angle where `Intensity_counts` equals 4095.0 and manually mask them.